



Studienjahr <i>Academic Year</i>	2023/2024		
Studiengruppe(n) <i>Study Group(s)</i>	BA, IM, LM		
Prüfung <i>Exam</i>	Klausur / <i>Written examination</i>		
Modul(e) <i>Module(s)</i>	Principles of Finance		
Zugelassene Hilfsmittel <i>Permitted aids</i>	Einsprachiges Wörterbuch, nicht-programmierbarer Taschenrechner / <i>monolingual dictionary, non-programmable calculator</i>		
Seiten Aufgabenstellung / Anlagen <i>Pages tasks / appendices</i>	16 Seiten inkl. Deckblatt / <i>16 pages incl. front page</i>	Formulary (3 pages)	
Prüfungsdatum / -zeit / -dauer <i>Examination date/time/duration</i>	24.06.2024	8 – 9.30 a.m.	90 Minuten <i>Minutes</i>
Prüfer <i>Examiner</i>	Hr. Prof. Prigge (modulverantwortlich)		

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Matrikelnummer <i>Matriculation number</i>			
Studiengruppe <i>Study Group</i>			
Anzahl zusätzlicher Seiten Papier <i>No. of additional sheets of paper</i>			
Klausurbeginn /-ende <i>Examination start / end</i>			

Punkteverteilung *Scores*

Aufgabe <i>Task</i>	1	2	3	4	Σ
Maximal <i>Maximum</i>	35	20	20	25	100
Erreicht <i>Achieved</i>					

Note <i>Grade:</i>	Prüfer <i>Examiner:</i>	Unterschrift Prüfer <i>Signature of the examiner:</i>

Punkte-Noten-Schema *Grading Scheme*

Punkte <i>Score</i>	100-95	94-90	89-85	84-80	79-75	74-70	69-65	64-60	59-55	54-50	49-0
Note <i>Grade</i>	1,0	1,3	1,7	2,0	2,3	2,7	3,0	3,3	3,7	4,0	n.b.



Aufgabe Task 1 Investment Decision Rules and WACC (35 Punkte Points)

As a member of the finance department of TipTop AG, you are involved in calculating the company's WACC as well as in preparing a major capital budgeting decision.

a) Your work starts with the WACC. Your colleagues provided the following data:

- cost of equity 10%
- cost of debt 5%
- market value of equity EUR100 million
- market value of debt EUR50 million
- corporate tax rate 33%

a1) Please calculate TipTop's WACC. (6 points)

a2) Please explain the tax advantage of debt in max. **three** sentences. (4 points)



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- a3) Corporate tax rates might change in the course of time. Such happened in the U.S., where the corporate tax rate was reduced from 35% to 21% in 2018. What was the effect on the after-tax cost of debt? No calculation required. **(3 points)**

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b) Turning to capital budgeting, you are part of the team that evaluates the new collection of flip-flop shoes that TipTop might introduce to the market next year. The whole project requires an initial investment outlay of EUR10 million. The following cash surpluses are expected over the next five years (see table; all amounts shown in EUR).

Year	1	2	3	4	5
Cash Flow	5,000,000	4,000,000	2,000,000	1,500,000	750,000

b1) The CFO asks you to calculate the NPV with the WACC you calculated under a1) as discount rate. Should TipTop AG realize the project? [If you did not calculate a WACC under a1), then use a WACC of 10%.] **(12 points)**



- b2) In b1), the company WACC of TipTop AG is applied to calculate the NPV and make the investment decision regarding the new flip-flop collection. Under which condition is that permissible? What should be done if the condition is not met? **(4 points)**

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- b3) How would you approach the decision about the flip-flop collection if TipTop disposes for the next fiscal year of several other positive NPV investment projects but cannot, due to a limited budget for investments, realize all of them? Please sketch briefly your proposal and calculate the relevant KPI for the flip-flop project. **(6 points)**

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(20 Punkte Points)

Rating	AAA	AA	A	BBB	BB
YTM	3.7%	3.9%	4.0%	4.7%	8.0%

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- c) How much total principal amount of these bonds must Company A issue to raise USD12 million today, assuming the bonds are AAA rated? (Because Company A cannot issue a fraction of a bond, assume that all fractions are rounded to the nearest whole number.) **(4 points)**

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[illegible]



Aufgabe Task 3 Stocks and Stock Market (20 Punkte Points)

- a) Triple-A AG forecasts EUR5.70 as earnings per share (EPS) for year-end. The company's CFO currently assumes earnings to remain constant for all future years. Triple-A's equity cost of capital is 12%.

- a1) As there are currently no promising new investment projects, the company has a payout rate of 100%. What is the current stock price? **(4 points)**

- a2) What is Triple-A's stock price if the CFO decides to retain 30% of earnings to reinvest in new investment projects with an expected return of 7%? Is the reinvestment decision reasonable? Explain briefly. **(8 points)**



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- b)
- b1) An insider from XYZ AG has just leaked an expected dividend increase of 10% for the running business year in a financial magazine. Officially, the increase will be announced a few weeks later during the annual meeting of the company. The stock price of XYZ AG immediately increases by 10% on the publication date of the leak in the financial magazine. Please comment briefly on the prevailing form of market efficiency here. **(5 points)**

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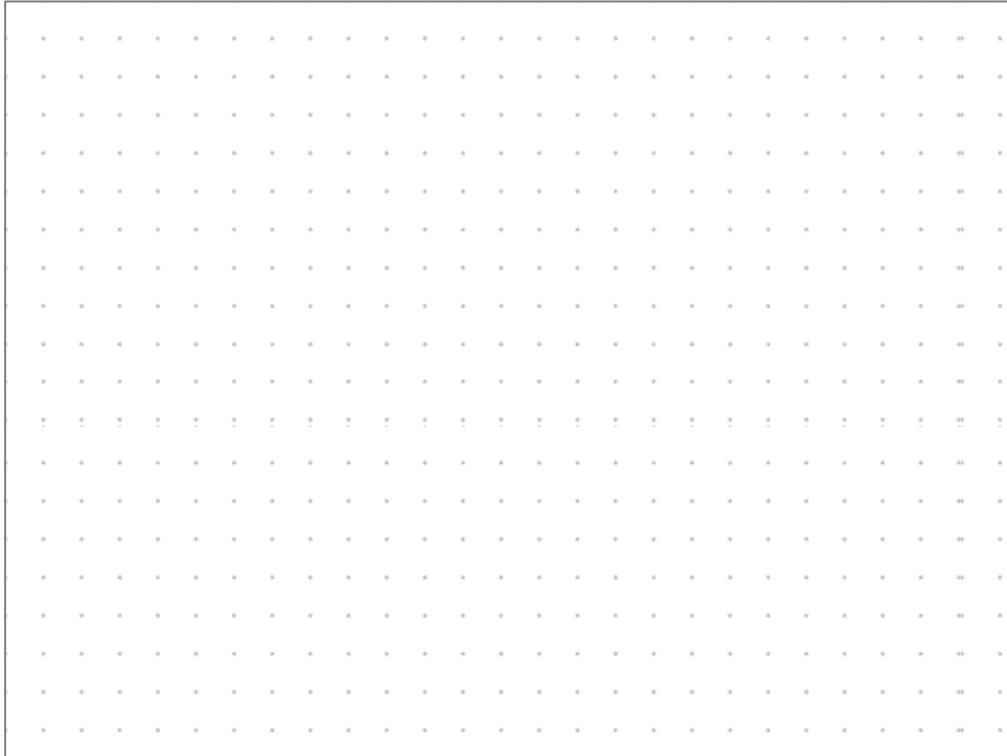
- b2) A few years later XYZ AG has a new ongoing business venture. In a few days the management team will hold a conference with investors to present a first outlook on the venture's prospects. If the venture is a failure, the severe amount of sunk costs will let the stock price crumble to EUR20. If the venture should be a promising success, it will lift XYZ AG to the top-class of their industry resulting in a share price of EUR85. The stock currently trades at EUR65. What is the expected probability of success for the venture deemed by financial markets (calculation required)?
(3 points)

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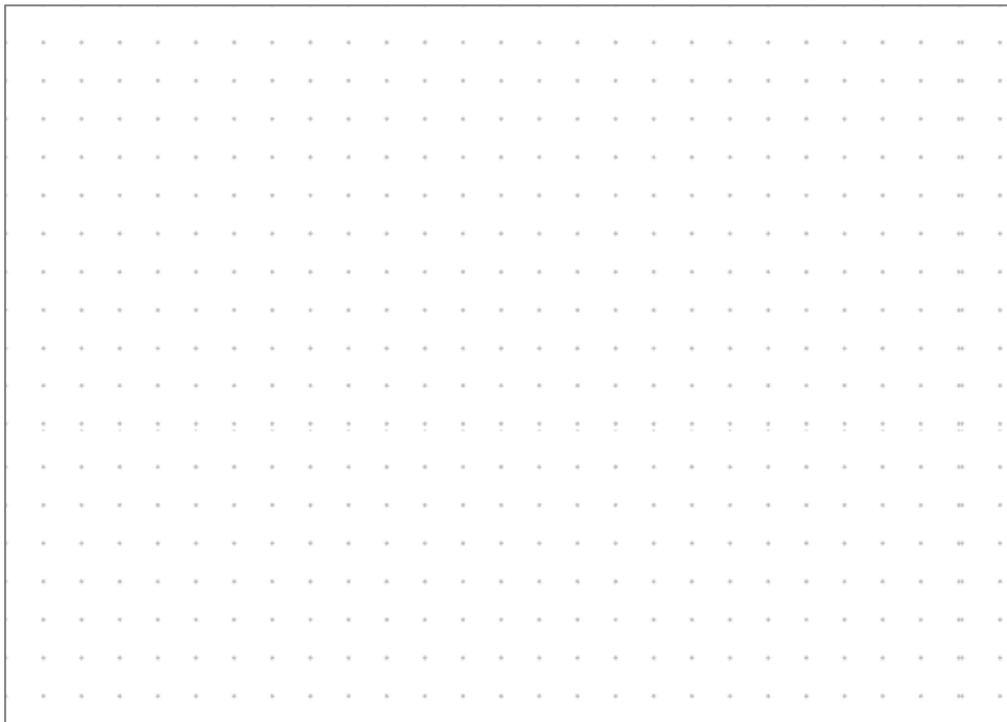
Aufgabe Task 4 Risk and Return

(25 Punkte Points)

- a) Please explain the difference between systematic and unsystematic risk.
(4 points)



- b) Please explain why there is no variable to represent the unsystematic risk in the SML formula of the CAPM. **(4 points)**

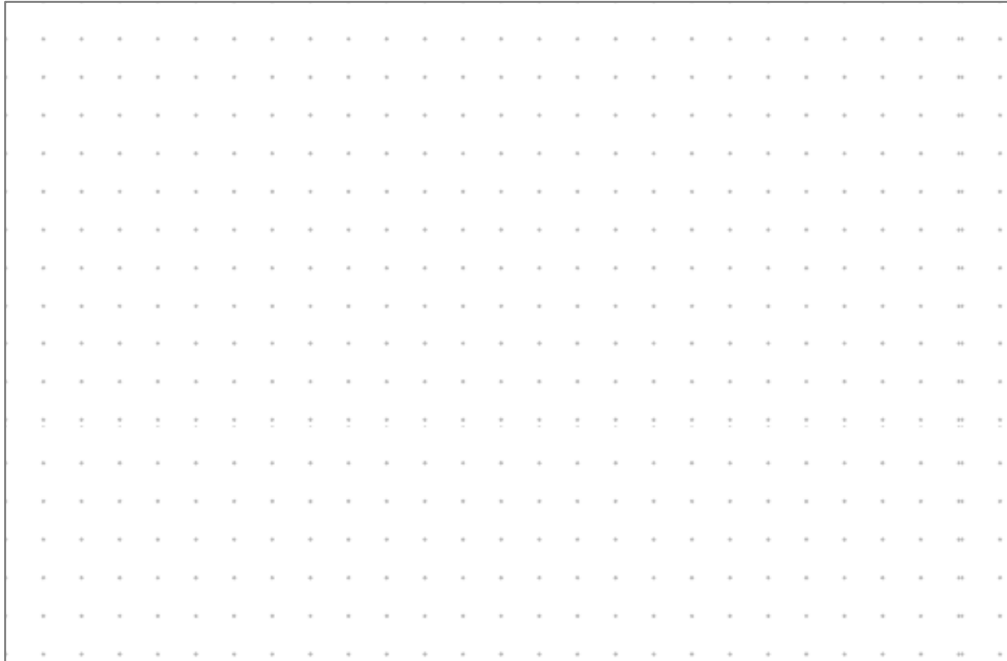


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- | | Recession | Normal | Boom | Expected Return | Standard Deviation |
|-----------|------------------|---------------|-------------|------------------------|---------------------------|
| TipTop AG | -9.00% | 8.00% | 21.00% | 6.67% | 12.28% |
| TopTip AG | 10.00% | 4.00% | -5.00% | 3.00% | 6.16% |

- [illegible]

- d3) Is the diversification effect caused by the addition of TopTip to your portfolio big? Please explain briefly. Is there a quick indicator to estimate the size of the diversification effect? **(3 points)**



Zusätzlicher Antwortbereich. Achten Sie bitte auf eine klare Zuordnung bzw. Nummerierung Ihrer Antworten. Additional answer area. Please make sure that your answers are clearly assigned or numbered.



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Formulary – Appendix to Exam Chapter 1

Future Value of a Cash Flow: $FV_n = C \times (1 + r) \times (1 + r) \times \dots \times (1 + r) = C \times (1 + r)^n$

Present Value of a Cash Flow: $PV = C \div (1 + r)^n = \frac{C}{(1 + r)^n}$

Net Present Value (NPV): compares the present value of cash inflows (benefits) to the present value of cash outflows (costs).

Present Value of a Perpetuity (first payment at time 1): $PV(C \text{ in perpetuity}) = \frac{C}{r}$

Present Value of an Annuity: $PV(\text{annuity of } C \text{ for } N \text{ periods with interest rate } r) = \frac{C}{r} \left(1 - \frac{1}{(1 + r)^N}\right)$

Future Value of an Annuity:

$$\begin{aligned} FV(\text{annuity}) &= PV \times (1 + r)^N \\ &= \frac{C}{r} \left(1 - \frac{1}{(1 + r)^N}\right) \times (1 + r)^N \\ &= C \times \frac{1}{r} ((1 + r)^N - 1) \end{aligned}$$

Present Value of a Growing Perpetuity: $PV(\text{growing perpetuity}) = \frac{C}{r - g}$

Present Value of a Growing Annuity: $PV = C \times \frac{1}{(r - g)} \left(1 - \left(\frac{1 + g}{(1 + r)}\right)^N\right)$

Approximate Internal Rate of Return (IRR) via Interpolation
(with $NPV_1 > 0$ and $NPV_2 < 0$): $r \approx r_1 - \frac{NPV_1}{NPV_2 - NPV_1} \cdot (r_2 - r_1)$

Chapter 2

Profitability Index: $\text{Profitability Index} = \frac{\text{Value Created}}{\text{Resource Consumed}} = \frac{NPV}{\text{Resource Consumed}}$

Chapter 3

Coupon Payment: $CPN = \frac{\text{Coupon Rate} \times \text{Face Value}}{\text{Number of Coupon Payments per Year}}$

Price of a Zero-Coupon Bond (FV=face value): $P = \frac{FV}{(1 + YTM)^n}$

Yield to Maturity of an n-Year Zero-Coupon Bond: $YTM = \left(\frac{FV}{P}\right)^{\frac{1}{n}} - 1$

Yield to Maturity of a Coupon Bond: discount rate y , that equates the PV of the bond's future cash flows with its price (=IRR): $P = CPN \times \frac{1}{YTM} \left(1 - \frac{1}{(1 + YTM)^N}\right) + \frac{FV}{(1 + YTM)^N}$

Chapter 4

Dividend-discount model (1 year): $P_0 = \left(\frac{\text{Div}_1 + P_1}{1 + r_E} \right)$

Total return, capital gain, dividend yield: $r_E = \frac{\text{Div}_1 + P_1}{P_0} - 1 = \underbrace{\frac{\text{Div}_1}{P_0}}_{\text{Dividend Yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{Capital Gain Rate}}$

Dividend-discount model (2 years): $P_0 = \frac{\text{Div}_1}{1 + r_E} + \frac{\text{Div}_2 + P_2}{(1 + r_E)^2}$

Dividend-discount model (n years): $P_0 = \frac{\text{Div}_1}{1 + r_E} + \frac{\text{Div}_2}{(1 + r_E)^2} + \dots + \frac{\text{Div}_N}{(1 + r_E)^N} + \frac{P_N}{(1 + r_E)^N}$

$$P_0 = \frac{\text{Div}_1}{1 + r_E} + \frac{\text{Div}_2}{(1 + r_E)^2} + \frac{\text{Div}_3}{(1 + r_E)^3} + \dots = \sum_{n=1}^{\infty} \frac{\text{Div}_n}{(1 + r_E)^n}$$

DDM, constant growth: $P_0 = \frac{\text{Div}_1}{r_E - g} \quad r_E = \frac{\text{Div}_1}{P_0} + g$

DDM, changing growth rates: $P_0 = \frac{\text{Div}_1}{1 + r_E} + \frac{\text{Div}_2}{(1 + r_E)^2} + \dots + \frac{\text{Div}_N}{(1 + r_E)^N} + \frac{1}{(1 + r_E)^N} \left(\frac{\text{Div}_{N+1}}{r_E - g} \right)$

Change in Earnings = New Investment × Return on New Investment

New Investment = Earnings × Retention Rate

Earnings Growth Rate = $\frac{\text{Change in Earnings}}{\text{Earnings}} = \text{Retention Rate} \times \text{Return on New Investment}$

$g = \text{Retention Rate} \times \text{Return on New Investment}$

Dividend: $\text{Div}_t = \underbrace{\frac{\text{Earnings}_t}{\text{Shares Outstanding}_t}}_{\text{EPS}_t} \times \text{Dividend Payout Rate}_t$

Chapter 5

Expected (Mean) Return: Calculated as a weighted average of the possible returns, where the weights correspond to the probabilities.

$$\text{Expected Return} = E[R] = \sum_R P_R \times R$$

Variance: The expected squared deviation from the mean.

$$\text{Var}(R) = E[(R - E[R])^2] = \sum_R P_R \times (R - E[R])^2$$

Variance Estimate Using Realized Returns:

$$\text{Var}(R) = \frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R})^2$$

Standard Deviation: The square root of the variance.

$$SD(R) = \sqrt{\text{Var}(R)}$$

$$\begin{aligned} \text{Portfolio rate of return} &= (\text{fraction of portfolio in asset 1} \times \text{rate of return on asset 1}) \\ &+ (\text{fraction of portfolio in asset 2} \times \text{rate of return on asset 2}) \\ &\dots \\ &+ (\text{fraction of portfolio in asset } t \times \text{rate of return on asset } t) \end{aligned}$$

$$\begin{aligned} \text{Portfolio beta} &= (\text{fraction of portfolio in stock 1} \times \beta \text{ of stock 1}) \\ &+ (\text{fraction of portfolio in stock 2} \times \beta \text{ of stock 2}) \\ &\dots \\ &+ (\text{fraction of portfolio in stock } t \times \beta \text{ of stock } t) \end{aligned}$$

$$\text{Risk premium on the market} = r_m - r_f$$

$$\text{Risk premium on any asset } i = r_i - r_f$$

$$\text{SML equation: } r = r_f + \beta(r_m - r_f)$$

Chapter 6

$$r_{\text{assets}} = \frac{\text{total income}}{\text{value of investments}} = \frac{(D \times r_{\text{debt}}) + (E \times r_{\text{equity}})}{V}$$

$$r_{\text{assets}} = \frac{D}{V} \times r_{\text{debt}} + \frac{E}{V} \times r_{\text{equity}}$$

$$\text{After-tax cost of debt} = r_{\text{debt}} \times (1 - T_c).$$

$$\text{WACC} = \left[\frac{D}{V} \times (1 - T_c) r_{\text{debt}} \right] + \left[\frac{E}{V} \times r_{\text{equity}} \right]$$